A 2-Week Practical Plan

FFT · PSD · POD · DMD · SPOD

Built around modal / spectral decomposition methods for unsteady shock-wave and boundary-layer-interaction data — tied to Endevco pressure-sensor time series, Schlieren image sequences, and ANSYS Fluent CFD snapshots from the CSBLI (Mach 6.58) study.

How to use this document

Each day lists: (1) the concept for the day, (2) a time budget split into theory/reading vs. hands-on coding, (3) verified links to lecture videos, papers, and code, and (4) a concrete exercise — first on a toy/synthetic signal, then, from Week 2 onward, applied directly to your own sensor/CFD/Schlieren data. Links were checked at the time this was written (August 2026); if any has moved, search the title/author given alongside it.

Suggested pace: **~1.5–2 hours/day, 6 days/week**, with one light consolidation day and one full rest/buffer day per week — sized for someone already running an active PhD pipeline (CFD + experimental + manuscript work) rather than a full-time course load. Compress or stretch as your week allows; the sequence (SVD → FFT/PSD → POD → DMD → SPOD) matters more than the exact dates.

Core spine of this plan

- **Textbook + free lectures:** Brunton & Kutz, *Data-Driven Science and Engineering* (free 2nd-ed. PDF) — companion site databookuw.com has a lecture video per section, code in MATLAB/Python, and homework.
- **Video lectures:** Steve Brunton's YouTube channel — youtube.com/@Eigensteve (channel homepage, for browsing; each day below links the specific playlist/video instead of the channel).
- **Code:** github.com/dynamicslab (official code for the book) and Nathan Kutz's course-lecture archive at faculty.washington.edu/kutz.
- **Fluids-specific review:** Taira et al., "Modal Analysis of Fluid Flows: An Overview" (*AIAA Journal*, 2017) — arXiv:1702.01453, and its sequel "...Applications and Outlook" (2019) — arXiv:1903.05750.

I'm confident these are real, currently-indexed sources since I pulled them from live search rather than memory, but I'd still recommend opening each link once at the start of Week 1 to confirm it resolves for you before relying on the schedule.

Day 0 — 15-minute setup checklist

- Python env: pip install numpy scipy matplotlib pydmd (add pyspod only if you plan to use it instead of the MATLAB SPOD code).
- MATLAB: confirm the Signal Processing Toolbox is licensed (needed for pwelch); clone spod_matlab into a scratch folder first, then into *03_Code/* once you're comfortable with it.
- Pick one small, real dataset now (e.g. a single 40-bar Endevco run, sensors excluding A14B14/A2) to reuse across all 14 days — consistency matters more than variety here.
- Open and skim (don't read fully) the Brunton & Kutz free PDF and the Taira 2017 overview once, just so the table of contents is familiar before Day 1.

Week-at-a-Glance

Day	Focus	Time	Output
1	Linear algebra refresher & SVD	1.5 h	SVD of a Schlieren-frame matrix
2	FFT theory & practice	1.5 h	FFT of synthetic + real pressure signal
3	PSD & Welch's method	1.5 h	PSD of Endevco 8530C-50 traces
4	POD theory	2 h	Derive POD from SVD by hand + notes
5	POD hands-on	2 h	POD of CFD snapshots / Schlieren sequence
6	Consolidation (light)	1 h	1-page summary: SVD→POD pipeline
7	Rest / buffer	—	Catch-up only if needed
8	DMD theory	1.5 h	Worked DMD algorithm by hand on toy data
9	DMD hands-on (PyDMD)	2 h	DMD of shock-oscillation time series
10	SPOD theory	1.5 h	Notes: POD vs DMD vs SPOD, when to use which
11	SPOD hands-on	2 h	SPOD of pressure array / CFD field
12	Comparative synthesis	1.5 h	Side-by-side mode comparison figure
13	Applied mini-project on CSBLI data	2 h	MATLAB step-style script (stepC-modal)
14	Wrap-up & write-up	1 h	Short internal note + next-steps list

Week 1 — Foundations: SVD, FFT, PSD, POD

Day 1 — Linear Algebra Refresher & the SVD

Suggested slot: Mon, ~1.5 h | **Time budget:** 45 min theory/video + 45 min coding

Learn (theory/reading):

- Why almost every method this fortnight (POD, DMD, SPOD) reduces to the SVD underneath.
- Geometric picture of SVD: rotation → scaling → rotation; interpretation of U, Σ, V.
- Economy vs. full SVD, rank truncation, and why truncation = 'denoising + compression'.

Materials & links:

- Video: Brunton, "Singular Value Decomposition (SVD)" playlist — YouTube playlist (linked directly from the book's own Ch. 1 page: databookuw.com, Ch. 1).
- Reading: Brunton & Kutz, *Data-Driven Science and Engineering*, Ch. 1 — free PDF (databookV2.pdf)
- Reference code (MATLAB/Python) for Ch. 1 examples: github.com/dynamicslab

Hands-on exercise:

In Python or MATLAB, take any single Schlieren frame (or a stack of ~50-100 frames reshaped into a snapshot matrix, columns = time) and compute its SVD. Plot the singular value spectrum (log scale) and reconstruct the frame using 5, 20, and 50 modes to see how quickly energy concentrates.

Applied to your CSBLI data:

Use frames from *04_Results/Schlieren/Selected_Frames* or a small chunk of your CFD export CSVs (e.g. one z-slice from *justPlate/40bar/exports*) as the snapshot matrix — this is exactly the data structure POD will need on Day 5.

Day 2 — FFT — Theory & Practice

Suggested slot: Tue, ~1.5 h | **Time budget:** 40 min theory + 50 min coding

Learn (theory/reading):

- DFT vs FFT (why FFT is fast: O(N log N) via divide-and-conquer).
- Sampling theorem, Nyquist frequency, aliasing — critical for your Endevco sampling rate.
- Windowing and spectral leakage (why a raw FFT of a finite-length signal lies to you).

Materials & links:

- Official tutorial: SciPy FFT tutorial
- Video: Brunton, "The Fast Fourier Transform Algorithm" — direct video (part of the Ch. 2 playlist linked from databookuw.com, Ch. 2).
- Background reading: Brunton & Kutz Ch. 2 (Fourier/Wavelet transforms) — databookV2.pdf

Hands-on exercise:

Generate a synthetic signal (e.g. two sine tones + noise) and take its FFT; identify the two peaks. Then apply numpy.fft.rfft to a single-channel segment of real Endevco pressure data and plot the one-sided amplitude spectrum. Try zero-padding and a Hann window and see how the peaks sharpen/blur.

Applied to your CSBLI data:

Run this directly on one excluded-sensor-free channel (recall A14B14 and A2 are excluded) from a 40-bar run to get a first look at dominant shock-oscillation frequencies before the more careful PSD treatment tomorrow.

Day 3 — Power Spectral Density & Welch's Method

Suggested slot: Wed, ~1.5 h | **Time budget:** 40 min theory + 50 min coding

Learn (theory/reading):

- Why a raw $|FFT|^2$ periodogram is noisy/inconsistent as an estimator.
- Welch's method: segmenting, windowing, overlapping, and averaging to reduce variance.
- Trade-offs: segment length vs. frequency resolution vs. variance (bias-variance for spectra).

Materials & links:

- Docs: [scipy.signal.welch reference]()
- Original method: P. Welch (1967), "The Use of the Fast Fourier Transform for the Estimation of Power Spectra" — I don't have a verified free-PDF link for this specific 1967 IEEE paper, so search the title directly via your institutional access.
- Practical framing (same Welch-based estimator SPOD will reuse next week): Schmidt & Colonius, "Guide to Spectral Proper Orthogonal Decomposition" — [open-access PDF, Caltech repository]()

Hands-on exercise:

Compute PSDs with scipy.signal.welch for 2-3 Endevco channels at a given fill pressure, varying nperseg (e.g. 512 vs 4096) to see the resolution/variance trade-off directly. Overlay PSDs across two fill pressures (e.g. 10 bar vs 40 bar) on one log-log plot.

Applied to your CSBLI data:

This gives you a first quantitative, defensible answer to "what are the dominant shock-unsteadiness frequencies at each fill pressure?" — worth keeping as a figure candidate for the manuscript's discussion of unsteadiness, separate from the GCI/validation work already in progress.

Day 4 — Proper Orthogonal Decomposition (POD) — Theory

Suggested slot: Thu, ~2 h | **Time budget:** 1 h theory/reading + 1 h worked derivation

Learn (theory/reading):

- POD as SVD applied to a mean-subtracted snapshot matrix; energy-optimal basis in the L^2 sense.
- Method of snapshots (Sirovich) for when spatial DOF ■ number of snapshots — relevant since your CFD fields have millions of cells but far fewer time snapshots.
- What POD modes physically represent vs. what they don't: energy ranking ≠ dynamical importance.

Materials & links:

- Taira et al. 2017, "Modal Analysis of Fluid Flows: An Overview", Sec. on POD — [arXiv:1702.01453]()
- Brunton & Kutz Ch. 1 (SVD) reframed as POD, plus Ch. 12 (if covered in your edition) — [databookV2.pdf]()
- Classic reference (cite in your manuscript if you use POD): Holmes, Lumley & Berkooz, *Turbulence, Coherent Structures, Dynamical Systems and Symmetry* (Cambridge, 1998) — I don't have a verified free-PDF link for this book; check your institutional library.

Hands-on exercise:

By hand (or in a short script), show that POD of a mean-subtracted snapshot matrix X is algebraically identical to SVD of X. Write a half-page note distinguishing POD (space-only, energy ranked) from what SPOD will give you next week (space-time, frequency ranked).

Applied to your CSBLI data:

Frame this note around your own decision point: your data includes both CFD snapshots (good spatial resolution, few snapshots) and 500 kHz-class pressure time series (many snapshots, sparse spatial points) — POD suits the former, SPOD the latter.

Day 5 — POD — Hands-On With Your Own Data

Suggested slot: Fri, ~2 h | **Time budget:** 20 min setup + 1 h 40 min coding

Learn (theory/reading):

- Practical POD pipeline: mean subtraction → snapshot matrix → SVD → mode energy plot → mode shapes.
- How many modes to keep (cumulative energy threshold, e.g. 90%/99%).

Materials & links:

- Same core references as Day 4; for code, adapt the SVD script from Day 1 rather than a new library.
- If you want a ready-made reference implementation style, browse github.com/dynamicslab for POD/SVD example notebooks.

Hands-on exercise:

Build a snapshot matrix from either (a) a sequence of Schlieren frames spanning one run, or (b) a spanwise CFD export set (e.g. the z_00–z_40 spanwise CSVs from *justPlate/40bar/exports*). Compute POD, plot the energy spectrum, and visualize the first 3-4 spatial modes.

Applied to your CSBLI data:

Check whether the leading POD modes pick out the conical shock foot / separation region location predicted by your analytical relief model (L■D/W relation) — a qualitative cross-check between the data-driven and analytical pictures.

Day 6 — Consolidation — Week 1 Recap

Suggested slot: Sat, ~1 h (light) | **Time budget:** 1 h, no new material

Learn (theory/reading):

- Re-read your own notes from Days 1-5; fix anything that didn't fully click.

Materials & links:

- No new links — revisit whichever Day 1-5 resource felt weakest.

Hands-on exercise:

Write a 1-page personal note: "SVD → FFT/PSD → POD" pipeline, in your own words, with one plot from each day pasted in. This is the artifact that will make Week 2 (DMD, SPOD) click faster since both build directly on this foundation.

Applied to your CSBLI data:

Optional: sanity-check your PSD peak frequencies (Day 3) against the POD mode structure (Day 5) — do the energetic POD modes oscillate near the dominant PSD frequency? This is the natural bridge into why SPOD exists.

Day 7 — Rest / Buffer Day

Suggested slot: Sun | **Time budget:** —

Learn (theory/reading):

- No scheduled learning — use only if you fell behind.

Materials & links:

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Hands-on exercise:

If you're on schedule, take the day off. If not, use it to finish Day 5's exercise.

Applied to your CSBLI data:

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Week 2 — Dynamics: DMD, SPOD, and Synthesis

Day 8 — Dynamic Mode Decomposition (DMD) — Theory

Suggested slot: Mon, ~1.5 h | **Time budget:** 1 h theory + 30 min worked example

Learn (theory/reading):

- DMD as a best-fit linear operator A between snapshot pairs (X, X'); eigendecomposition of A gives frequency + growth/decay per mode.
- Connection to the Koopman operator — DMD as a finite-dimensional Koopman approximation.
- Key contrast with POD: DMD modes are single-frequency and ranked by dynamical relevance, not energy.

Materials & links:

- Book: Kutz, Brunton, Brunton & Proctor, *Dynamic Mode Decomposition: Data-Driven Modeling of Complex Systems* (SIAM, 2016) — companion site dmdbook.com (has the fluid-dynamics chapter and code index).
- Foundational theory paper: Tu, Rowley, Luchtenburg, Brunton & Kutz, "On Dynamic Mode Decomposition: Theory and Applications", *J. Comput. Dyn.* 1(2), 2014 — arXiv:1312.0041.
- Video: Brunton, "Dynamic Mode Decomposition (Overview)" — direct video and "Dynamic Mode Decomposition (Theory)" — direct video.

Hands-on exercise:

On a small synthetic dataset (e.g. two spatial patterns each oscillating at a distinct known frequency, summed), hand-code the basic DMD algorithm (SVD of X, build Ã, eigendecompose, reconstruct DMD modes) and confirm you recover both known frequencies.

Applied to your CSBLI data:

Keep this toy case as your 'unit test' — you'll rerun it against PyDMD tomorrow to confirm your library output matches your own from-scratch implementation.

Day 9 — DMD — Hands-On With PyDMD

Suggested slot: Tue, ~2 h | **Time budget:** 20 min setup + 1 h 40 min coding

Learn (theory/reading):

- Practical DMD pipeline using a maintained library rather than from-scratch code.
- Reading the DMD eigenvalue plot: unit circle = neutral, inside = decaying, outside = growing.

Materials & links:

- Library: PyDMD (github.com/PyDMD/PyDMD) — install via pip install pydmd; repo includes tutorial notebooks.
- Companion paper/tutorial: Demo, Tezzele & Rozza, "PyDMD: Python Dynamic Mode Decomposition" (JOSS, 2018) — referenced in the PyDMD README above.

Hands-on exercise:

Run PyDMD on the same snapshot matrix you built for POD on Day 5 (Schlieren sequence or spanwise CFD slices). Plot the DMD eigenvalue spectrum and the frequency/growth-rate of the top 5 modes; compare mode shapes against your Day 5 POD modes.

Applied to your CSBLI data:

Apply DMD to a time-resolved pressure-sensor array (if you have simultaneous multi-channel Endevco data) to extract the dominant oscillation frequency and its growth/decay behavior — directly relevant to characterizing shock unsteadiness rather than just its mean structure.

Day 10 — Spectral POD (SPOD) — Theory

Suggested slot: Wed, ~1.5 h | **Time budget:** 1 h reading + 30 min notes

Learn (theory/reading):

- SPOD as space-time POD for statistically stationary data: each SPOD mode oscillates at a single frequency AND is energy-ranked within that frequency — a genuine hybrid of POD and DMD.
- Relationship to DMD and resolvent analysis (SPOD modes converge to DMD modes under certain limits).
- Why SPOD suits stochastic/turbulent unsteadiness (like SWBLI shock motion) better than plain POD or DMD alone.

Materials & links:

- Foundational paper: Towne, Schmidt & Colonius, "Spectral Proper Orthogonal Decomposition and Its Relationship to Dynamic Mode Decomposition and Resolvent Analysis", *J. Fluid Mech.* 847 (2018) — arXiv preprint (1708.04393)
- Practical guide: Schmidt & Colonius, "Guide to Spectral Proper Orthogonal Decomposition", *AIAA Journal* 58 (2020) — open-access PDF, Caltech repository

Hands-on exercise:

Write a short comparison table in your own words: POD vs. DMD vs. SPOD — what each optimizes for, what input data shape each needs, and which one you'd reach for given (a) a single CFD snapshot set, (b) a long stationary pressure time series, (c) a short transient event.

Applied to your CSBLI data:

Decide, in writing, which of your CSBLI datasets is genuinely 'statistically stationary' (a precondition for SPOD) — likely your longer Endevco pressure records at fixed fill pressure, less so a short transient CFD convergence sequence.

Day 11 — SPOD — Hands-On

Suggested slot: Thu, ~2 h | **Time budget:** 20 min setup + 1 h 40 min coding

Learn (theory/reading):

- Practical SPOD pipeline: block Welch segmentation of the snapshot sequence, per-frequency cross-spectral matrix, eigendecomposition per frequency.

Materials & links:

- MATLAB reference implementation: github.com/SpectralPOD/spod_matlab (also mirrored on MathWorks File Exchange: SPOD (FileExchange #65683)).
- Since your existing pipeline is MATLAB-based (stepC0-C7 / step0-step7), spod_matlab is the more natural fit than a Python port — it drops into your existing *03_Code/* structure.
- Python alternative, if you ever want it (e.g. for a portable analysis script outside the MATLAB pipeline): PySPOD (github.com/MathEXLab/PySPOD) — also supports MPI-parallel SPOD for larger CFD snapshot sets.

Hands-on exercise:

Run SPOD on either your multi-channel Endevco pressure array (time as the fast axis, sensor location as the spatial axis) or a time-resolved CFD monitor sequence. Plot the SPOD eigenvalue spectrum vs. frequency (the characteristic 'SPOD spectrum' plot) and the leading mode shape at the dominant frequency identified in Day 3's PSD.

Applied to your CSBLI data:

This is the step most directly reusable for the manuscript: an SPOD spectrum of the shock/BL interaction unsteadiness, with the leading low-rank mode shape at the dominant frequency, is a standard and expected figure type in SWBLI literature.

Day 12 — Comparative Synthesis: POD vs. DMD vs. SPOD

Suggested slot: Fri, ~1.5 h | **Time budget:** 1.5 h, mostly hands-on

Learn (theory/reading):

- Consolidating all three decompositions applied to the same dataset side by side.

Materials & links:

- Taira et al. 2019, "Modal Analysis of Fluid Flows: Applications and Outlook" — [arXiv:1903.05750](https://arxiv.org/abs/1903.05750) (worked application-level comparisons across method types).

Hands-on exercise:

Produce one figure with three panels: the leading POD mode (Day 5), the leading DMD mode at its natural frequency (Day 9), and the leading SPOD mode at the same frequency (Day 11), all on the same dataset and color scale. Write 3-4 sentences on where they agree/disagree and why.

Applied to your CSBLI data:

This comparison figure is a strong, defensible way to justify to reviewers *why* you chose whichever method(s) you ultimately use in the manuscript, rather than presenting just one method's output without context.

Day 13 — Applied Mini-Project: Integrate Into Your CSBLI Pipeline

Suggested slot: Sat, ~2 h | **Time budget:** 2 h coding

Learn (theory/reading):

- Turning this fortnight's scripts into a reusable pipeline step.

Materials & links:

- No new external material — this is integration work using everything above.

Hands-on exercise:

Write a single MATLAB script, following your existing pipeline convention (full file path in every fprintf/disp line, per your established convention) that takes a snapshot matrix and outputs PSD, POD, DMD, and SPOD results plus standard diagnostic plots for any of your four fill pressures (10/20/30/40 bar).

Applied to your CSBLI data:

Slot this in as a new step (e.g. *stepC-modal.m* alongside stepC0-C7) so future-you can rerun modal analysis on any new CFD case or experimental run without rebuilding the pipeline from scratch.

Day 14 — Wrap-Up & Write-Up

Suggested slot: Sun, ~1 h | **Time budget:** 1 h

Learn (theory/reading):

- Consolidating two weeks of notes into something citable and reusable.

Materials & links:

- Revisit Taira et al. 2017/2019 and the Schmidt & Colonius SPOD guide for correct terminology/citation format when you write this up.

Hands-on exercise:

Write a half-to-one-page internal summary: what each method showed on your data, which method(s) you plan to carry into the manuscript, and a short list of open questions (e.g. sensitivity of SPOD block length, number of POD modes needed for convergence).

Applied to your CSBLI data:

File this alongside your existing manuscript corrections log (C1-C12) as a reference note for when modal-analysis results get folded into the Physics of Fluids / JFM submission.

Appendix A — Software Setup

- Python: numpy, scipy (FFT, Welch), matplotlib, pydmd (pip install pydmd).
- MATLAB: base + Signal Processing Toolbox (for pwelch); spod_matlab cloned into your *03_Code/* tree.
- No GPU or special hardware needed — all methods here are dense linear algebra on snapshot matrices sized for a laptop/workstation, well within what you're already using for the MATLAB pipeline.

Appendix B — Full Reference List

1 Brunton, S. L. & Kutz, J. N. *Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control*, 2nd ed. Cambridge University Press, 2022. Free PDF: databookuw.com/databookV2.pdf · Course site: databookuw.com · Code: github.com/dynamicslab

2 Kutz, J. N., Brunton, S. L., Brunton, B. W. & Proctor, J. L. *Dynamic Mode Decomposition: Data-Driven Modeling of Complex Systems*. SIAM, 2016. Companion site: dmdbook.com

3 Brunton, S. L. (YouTube, "@Eigensteve"). Channel homepage (for browsing all playlists): youtube.com/@Eigensteve — specific SVD/FFT/DMD videos and playlists used in this plan are linked in-text on their respective days above.

4 Kutz, J. N. Graduate course lecture archive (AMATH 563, AMATH 584, and Data-Driven Science and Engineering courses): faculty.washington.edu/kutz

5 Taira, K., Brunton, S. L., Dawson, S. T. M., Rowley, C. W., Colonius, T., McKeon, B. J., Schmidt, O. T., Gordeyev, S., Theofilis, V. & Ukeiley, L. S. "Modal Analysis of Fluid Flows: An Overview." *AIAA Journal* 55(12), 4013–4041, 2017. arXiv:1702.01453

6 Taira, K., Hemati, M. S., Brunton, S. L., Sun, Y., Duraisamy, K., Bagheri, S., Dawson, S. T. M. & Yeh, C.-A. "Modal Analysis of Fluid Flows: Applications and Outlook." *AIAA Journal* 58(3), 998–1022, 2020. arXiv:1903.05750

7 Towne, A., Schmidt, O. T. & Colonius, T. "Spectral Proper Orthogonal Decomposition and Its Relationship to Dynamic Mode Decomposition and Resolvent Analysis." *Journal of Fluid Mechanics* 847, 821–867, 2018. arXiv preprint

8 Schmidt, O. T. & Colonius, T. "Guide to Spectral Proper Orthogonal Decomposition." *AIAA Journal* 58(3), 1023–1033, 2020. Open-access PDF (Caltech repository)

9 SpectralPOD (Schmidt, Towne, Colonius). SPOD MATLAB implementation: github.com/SpectralPOD/spod_matlab · also on MATLAB File Exchange

10 Tu, J. H., Rowley, C. W., Luchtenburg, D. M., Brunton, S. L. & Kutz, J. N. "On Dynamic Mode Decomposition: Theory and Applications." *Journal of Computational Dynamics* 1(2), 391–421, 2014. arXiv:1312.0041

11 PyDMD/PyDMD. Python Dynamic Mode Decomposition library: github.com/PyDMD/PyDMD

12 MathEXLab/PySPOD (Mengaldo, Maulik, Rogowski, Yeung, Schmidt et al.). Parallel Python SPOD package: github.com/MathEXLab/PySPOD

13 SciPy documentation. FFT tutorial: docs.scipy.org/doc/scipy/tutorial/fft.html · Welch's method reference: scipy.signal.welch

Note on sourcing: every link above was pulled from a live web search while preparing this document rather than recalled from memory, specifically to avoid citing a plausible-looking but broken or fabricated URL. Two items are flagged in-text above as real, well-known references for which I did not find a verified free-PDF link: the original 1967 Welch paper (Day 3) and the Holmes/Lumley/Berkooz POD textbook (Day 4) — both are standard citations you'll likely have institutional access to; search the title/author directly rather than trusting a link I didn't check.